MILITARY SPECIFICATION

LATHE, ENGINE, 13-INCH SWING

FLOOR MOUNTED WITH CABINET BASE

This specification is approved for use by the US Army Armament, Munitions, and Chemical Command (Engineering Support Directorate, Tool and Equipment Engineering Division, Standardization, Specifications and Packaging Branch), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a 13-inch swing, cabinet mounted, engine lathe with the overall physical dimension limitations of 68 inches maximum length, 38 inches maximum width and 52 inches maximum height (all dimensions include the cabinet base). The lathe shall be no less than 25 inches between centers and shall have machining capabilities in the English inch system (US) and the Metric (SI) system of measurement.

2 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions, and Chemical Command, ATTN: DRSMC-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3416

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MIL-P-116 - Preservation-Packaging, Methods Of
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging Of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-129 - Marking For Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 79 - Electrical Standards for Metalworking Machine Tools

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B5.1 - T-Slots-Their Bolts, Nuts, Tongues, and Cutters
B5.8 - Chucks and Chuck Jaws
B5.9 - Spindle Noses for Tool Room Lathes, Engine Lathes, Turret Lathes, And Automatic Lathes
B5.10 - Machine Taper; Self-Holding and Steep Taper Series
B5.16 - Accuracy of Engine and Tool Room Lathes

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E380 - Standard for Metric Practice
D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

201.02 - Tooth proportions for coarse-pitch involute spur gears
390.03 - Gear handbook volume 1, Gear classification materials and measuring methods for unassemblied gears
600.01 - Standard for metric usage

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(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

Standard 1000 - SI units and recommendations for the use of their multiples and certain other units.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete lathe for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.1.1 Safety and health requirements. The machine shall be equipped with covers, guards, or other suitable safety devices to protect personnel from hazardous moving components of the machine and from electrical equipment, that can be guarded without interference with normal machine functions or operator performance. Guards or covers for mechanisms that require frequent operator attention such as for setting adjustments or maintenance shall be readily removable. The machine shall comply with the standards promulgated under OSHA that are applicable to the machine itself. The lathe, when in operation, shall not exceed a noise level of 80 decibels (measured four feet from the machine).

3.1.2 Design. The lathe and its accessories shall be new and one of the manufacturer's current models. The machine shall meet or exceed the requirements specified herein and shall be designed and constructed for convenient and safe operation. All parts subject to wear, breakage or distortion shall be accessible for adjustment, repair and replacement. The lathe shall meet or exceed the performance and accuracy requirements specified herein.

3.1.3 Construction. The lathe shall be of strong and durable construction and securely mounted on a rigid metal cabinet base. The lathe specified herein is intended for mobile field travel and will be installed on a truck bed. The lathe, mounted on the cabinet base, shall be capable of withstanding stresses and strains resulting from the floor movement associated with the described installation without encountering adverse affects. The headstock unit of the lathe shall be powered by a main drive motor, connected to the headstock unit, by either an all gear drive assembly or by the means of V-belt(s). The main drive motor shall be enclosed within the end gear train guard, mounted on the back of the lathe and guarded or contained in

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the cabinet base beneath the lathe. All parts and materials used in the construction of the lathe and its components shall be new and free of defects. The lathe shall be complete and ready for operation after initial set-up procedures and power application.

3.1.3.1 Cabinet base. The steel cabinet base shall be constructed sufficiently strong and rigid to support the lathe in a firm and stationary manner during lathe operations. The cabinet base shall be capable of withstanding the stresses and strains associated with mobile field travel explained in the preceeding paragraph. The cabinet shall be constructed with a chip pan and designed for floor mounting. Tool storage shall be provided in the cabinet to accommodate the carrying of tools, equipment, attachments, accessories and wrenches furnished with the lathe. Storage areas may be either drawers or compartments with shelves, or both. All drawers or compartment doors including, if required, an underneath main drive motor compartment, shall be provided with locks. The same key shall operate all locks and not less than three (3) keys shall be furnished. Inserts or liners shall be durable and fastened in the drawers or on the shelves to securely retain all tools, equipment, attachments, accessories and wrenches during mobile field travel.

3.1.4 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. No process such as, welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.1.5 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where included in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.1.6 Fastening devices. All screws, pins, bolts, and similar fasteners, used internally, shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.1.7 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.1.8 Threads. All threaded parts shall conform to Federal Standard H28 and applicable "Detailed Standard" section referenced therein.

3.1.9 Gears. All gears and pinions used in the machine or its components shall be fabricated in the English (US) system or the Metric (SI) system of measurement. English gears shall conform to AGMA 390.03. Spur gears shall conform to AGMA 201.02. Metric gears shall comply with the International System of Units (SI), International Standards Organization (ISO) Standard 1000, American Society of Testing Materials ASTM E 380, and American Gear Manufacturers Association AGMA 600.01.

3.1.10 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with proper lubrication to assume that no metal to metal contact exists between any rotating, sliding, or moving parts. The lubrication reservoir(s) shall have means for checking and determining the proper oil level. All oil holes, grease fittings, and filler caps shall be easily accessible.

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3.1.11 Interchangeability. All replacement parts shall be functionally interchangeable without causing any adverse affects on the lathe's operation or its accuracy.

3.1.12 Painting. The lathe shall be painted to a quality that is commensurate with a good commercial practice used by other manufacturers of industrial engine lathes. The color shall be in accordance with the manufacturer's recognized commercial practice.

3.2 Sizes and capacities. Unless otherwise specified (see 6.2.1) the lathe shall meet the sizes and capacities in accordance with Table I.

TABLE I. Sizes and capacities

(Dimensions are shown in inches unless otherwise stated)	
Maximum overall length (Including cabinet)	Not more than 68
Maximum overall width (Including cabinet and tapes attachment mounted)	Not more than 38
Maximum overall height (Including cabinet)	Not more than 52
Swing over bed and carriage wings	Not less than 13
Swing over cross slide	Not less than 7-3/4
Distance between centers	Not less than 25
Hole thru headstock spindle	Not less than 1-3/8
Spindle nose, size and type	4"-D1 Camlock
Number of spindle speed changes	12 or variable
Spindle speed range, Revolutions per minute (RPM)	55 or less to 1800 or more
Number of thread and feed changes	35 or more
Threads per inch	4 or less to 56 or more
Longitudinal Feed Range, inches per revolution (IPR)	0.001 or less to .035 or more
Cross feed range (IPR)	0.0005 or less to 0.020 or more
Number of thread changes, metric	20 or more
Thread range, lead in millimeters (MM)	0.2 or less to 14.0 or more
Cross slide travel	Not less than 7
Compound rest travel	Not less than 3-1/8
Tool post capacity	1/2 by 1 1/8
Lathe centers, ANSI taper size	No. 3
Tailstock spindle diameter	Not less than 1-9/16
Tailstock spindle travel	Not less than 4
Taper attachment, degrees	Not less than 16
Length of taper cut at one setting	Not less than 9
Chuck, 4-jaw, independent, diameter	Not less than 10
Chuck, 3-jaw, universal, diameter	Not less than 8
Drive plate, diameter	Not less than 6-1/2
Face plate, large diameter	Not less than 12
Steady rest capacity	3/8 to 3
Follow rest capacity	3/8 to 2 1/2 inches
Main drive motor horsepower	Not less than 1 1/2 HP

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3.2.1 Machine accuracy. The lathe shall meet or exceed all accuracy tests as set forth in ANSI B5.16 for 12-inch to 18-inch size engine lathes.

3.3 Functional performance and characteristics.

3.3.1 Bed. The lathe bed shall have integral cross ribs, and shall be stress relieved during manufacture to provide stability. The bed shall have two V-ways and two flat ways or three V-ways and one flat way for supporting and guiding the carriage and tailstock. All bed way surfaces upon which the carriage and tailstock slides shall be hardened to not less than Rockwell C-45.

3.3.2 Headstock. The headstock shall be either manual shift or the infinitely variable speed type (manufacturer's option). The spindle shall be operable in both forward and reverse directions of rotation. The headstock shall be constructed to rigidly support the spindle and gear shafts in a secure manner to prevent deflection that would affect work part accuracy under operating and full power cutting loads. Spindle drive gears shall have a surface hardness of not less than Rockwell C-48. Easy access to the drive mechanism shall be provided. During set-up operation the spindle shall be free to rotate manually, and there shall be provision for retaining the spindle in any desired position to facilitate the mounting or operations of chucks, face plates and other spindle mounted attachments. Headstocks which require a spindle locking device shall have an interlock safety mechanism to prevent power application while the spindle is locked in order to prevent damage. Headstocks, if belt driven, shall have means for changing, adjusting and maintaining proper belt tension without losing spindle alignment or adjustment. Lubrication of the headstock components shall be by any system, or combination of lubricating systems that provides a positive means of delivering oil to bearings (except sealed type), gears and sliding or rotating parts that require continuous lubrication. Splash systems, if used, shall have a provision for insuring adequate lubrication of all bearings and parts requiring lubrication. Gearing for the transmission shall run in a bath of oil with provisions for checking and maintaining the oil level in the transmission. Speed changes for a manual shift headstock shall be made by setting gear shift levers on the headstock, as indicated on a conveniently located plate, showing the speed settings for shifting gears or clutches to obtain the required speeds. Speeds shall be in an approximate geometric progression. The variable speed headstock shall provide infinitely variable spindle speed selections for all speeds within the range of the lathe. Speed selections shall be made by suitable controls on the headstock end of the lathe and a speed indicator shall be provided to show the actual revolutions per minute of the spindle. It shall be possible to make speed adjustments while performing cutting operations so that the full/speed ratio will be obtainable under various cutting conditions. The headstock spindle, on variable speed type headstocks, shall start and turn without sticking or jerking at any preselected RPM.

3.3.3 Headstock spindle. The spindle nose shall be a 4-Inch D1 Camlock and shall conform to Table I and ANSI B5.9 as applicable. The inner and outer surfaces of the spindle nose on which work holding devices are mounted shall have a surface hardness of not less than Rockwell C-40. The spindle and spindle drive gear shafts shall be mounted in precision type ball or roller bearings or replaceable sleeve type bearings. If ball or roller bearings are used the radial and axial movement of the spindle shall not exceed 0.0004 at the spindle nose. If sleeve type bearings are used they shall be mounted in bearing supports and line bored for accurate alignment, and shall have a surface finish not rougher than 20 microinches.

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3.3.4 Carriage. The carriage shall be accurately fitted to the mating bedway surfaces providing smooth, constant feed motion under power and shall have adjustable gibs to permit compensation for wear. A carriage clamping device shall securely lock the carriage in any position along the bed when cross feeding. A carriage reversing mechanism shall allow feed movement in both directions without changing direction of spindle rotation. The reversing control shall be easily operated and positive acting, located within easy reach from the operator's normal work position. Way wipers shall be provided on the carriage wings to prevent chips and abrasives from entering between the carriage and bedways. The carriage shall be provided with necessary means for attaching accessories such as follow rests and a taper attachment.

3.3.5 Cross slide and compound rest. The cross slide shall be fitted to the carriage. A full length, adjustable gib shall be provided to compensate for wear between the slide and carriage ways. The compound slide shall consist of a swivel base and a top slide. The swivel base shall be attached to the cross slide by means of an accurately fitted swivel bearing and shall be arranged to allow the top slide to be pivoted 360 degrees with means for securely locking the swivel in the desired position. The swivel base shall be graduated in one degree increments from 0 to 90 degrees in both directions from the zero point. The top slide base shall be suitably marked to facilitate accurate angular settings and when set to the zero degree, the transverse movement of the slide shall be perpendicular to the longitudinal movement of the carriage. The cross feed screw threads shall be protected from chips and other foreign matter by the construction of the cross slide or by a suitable guard. The top slide shall have a standard T-slot in accordance with ANSI B5.1 for mounting a tool post.

3.3.6 Cross feed screws. The cross slide and top slide feed screw threads shall be acme or square form. Feed dials shall show tool movement or stock removal in increments of 0.001 inch and 0.02 mm.

3.3.7 Apron. The apron shall be double wall construction providing inboard and outboard supports for gear shafts and studs. Gearing in the apron shall provide hand and power longitudinal and cross feeds. Feeds shall be engaged by means of a clutch operated from the front of the apron. The clutch shall be a safety type to protect the feed mechanism against damage. A selector lever shall be provided on the front of the apron for engaging the cross and longitudinal feeds. An interlock shall be incorporated in the apron to prevent engagement of the feed and half or 2-piece split nut at the same time. The apron shall have an oil reservoir with sight glass and a self-contained lubrication system.

3.3.8 Power feed. The longitudinal and cross power feeds shall be from a feed rod and leadscrew, through a geared feed mechanism within the apron. The longitudinal feed shall be by a rack mounted on the underside of the front bed way, and meshing pinion on the apron. The rack and pinion tooth hardness difference, or use of different metals shall be designed to impart most wear to the component most inexpensively replaced. The feed and threading selections shall be made by shifting gears in an enclosed, quick change gear box. The gears shall provide the full range of feed and thread selections by dial or lever type gear shift selectors. Quick change transposing gears shall be furnished, if required, to produce English or Metric threads when the dual capability is not provided through the lathe's integral gearing. Lubrication shall be provided for the quick change gear box and gearing. Safety clutches or other suitable devices shall protect the feed mechanism from damage by accidental overload. A feed and thread chart shall be conveniently located to show control settings.

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3.3.9 Tailstock. The tailstock shall be accurately fitted to the bed ways and aligned with the headstock spindle. The tailstock shall be marked to indicate a set-over of at least one-half inch each side of zero as aligned with the headstock spindle centerline. A sensitive adjusting screw shall be provided to facilitate set-over alignment. The spindle taper shall be in accordance with ANSI B5.10 for self-holding tapers and shall eject tools when fully retracted. The spindle shall have a hardness of not less than Rockwell C-38 and shall be graduated in increments not greater than 1/16 inch and 1 mm for measuring spindle extension. The spindle shall be easily extended and retracted by a handwheel without jamming when run to both limits of travel. The tailstock shall be easily moved along the bed ways and have a quick acting clamp for locking the tailstock to the bed and for locking the spindle in any extended position without deflection. The spindle mechanism and tailstock guide ways shall be provided with means for lubrication.

3.3.10 Data plates. Metal plates, legibly and permanently marked in the English language to indicate proper control settings for each feed or speed obtainable, shall be permanently attached to the gear box and headstock. All functional controls shall be identified.

3.3.11 Electrical system. The electrical system shall conform to the National Fire Protection Association (NFPA) Electrical Standard for Metalworking Machine Tools No. 79. The lathe shall operate on 230/460 volt, 3 phase, 60 Hz alternating current (AC) power and shall be initially wired for operation on the 230 voltage. An identified terminal shall be provided that is suitable for connection of the size of grounding conductor specified in Table 240-A of the NFPA Standard. An emergency off switch, disconnecting all incoming power to the lathe, shall be provided in the front area of the lathe or cabinet base convenient to the operator's position. An interlock shall be provided to prevent power application while the lathe spindle is locked.

3.3.12 Motor. The main drive motor shall be dual voltage 230/460, 60 Hz, 3 phase AC and be not less than 1 1/2 horsepower. The motor shall contain permanently sealed ball or roller bearings, rated for continuous duty operation, have class B insulation, meet the requirements of a drip-proof enclosure and shall conform to applicable requirements of CC-M-1807.

3.3.13 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to the lathe. If a chart is furnished it shall be placed in a transparent plastic folder or laminated between clear plastic sheets, permanently sealed, with suitable means for mounting on the lathe. The following information shall be provided on the chart or plate:

Points of Application
Service Interval
Type of Lubricant
Viscosity
Federal or Military Specification Number(s), if available

3.3.14 Nameplate. A corrosion resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated provided the entry for each such caption is clear as to its identity.

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Nomenclature
Manufacturer's Name
Manufacturer's Model Designation
Manufacturer's Serial Number
Power Input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number
Date of Manufacture

3.4 Equipment. The following attachments and accessories shall be supplied with each lathe.

3.4.1 Thread chasing indicator. The dial indicator, engaging the leadscrew, shall be attached to the carriage apron. It shall be graduated to indicate half, or 2-piece split, nut engaging position when threading inch threads.

3.4.2 Drive plate. The drive plate shall be constructed from cast iron or semi-steel and shall securely mount on the spindle nose. The plate shall have not less than two radial slots for driving a lathe dog. The drive plate shall be capable of turning the maximum lathe speed without exceeding safety level recommendations.

3.4.3 Lathe centers. Two lathe centers shall be provided (one headstock and one tailstock). The centers shall be hardened to Rockwell C-62 to C-68 and ground to an accuracy within 0.0002 inch total indicator reading for concentricity. The centers shall conform to ANSI B5.10.

3.4.4 Center sleeve. One headstock center sleeve may be provided for adapting the taper in the spindle nose to the taper of the headstock center supplied. Concentricity shall be within 0.0002 inch total indicator reading. Tapers shall be in accordance with ANSI B5.10 for self-holding tapers.

3.4.5 Tool post. The tool post shall be round, single screw type of forged steel, or machined from a solid steel bar, and shall be complete with rocker support and ring.

3.4.6 Taper attachment. The taper attachment shall be a telescoping screw type for mounting in any working position on the rear of the lathe bed. It shall be graduated for setting in both degrees and inches per foot. Turning capacity shall be at least nine inches in length in one setting and 16 inches included angle. The attachment shall not interfere with normal lathe operation when not in use.

3.4.7 Face plate, large. The face plate shall be 12 inches nominal in diameter and shall be constructed of cast iron or semi-steel. The plate shall be of a ribbed type construction and shall be capable of turning the maximum lathe speed without exceeding safety level recommendations. The plate shall have four or more T-slots or through slots and comply with ANSI B5.1. The plate shall securely mount to the spindle nose.

3.4.8 Indicator carriage stop. The carriage stop shall have an indicator dial or micrometer graduated in 0.001 inch increments or less, and an adjustable stop rod for accurate carriage positioning. Dual measuring capability is required and two interchangeable indicators shall be provided; one for indicating in the US inch units and one for SI (metric) units.

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3.4.9 Chuck, 4-jaw, independent. The chuck body shall be iron alloy (semi-steel) and shall fit the spindle nose without adapters other than as part of the chuck. Four reversible step jaws with adjusting screws and wrench shall be included. All jaws and adjusting screws shall be hardened to a Rockwell scale sufficient for their intended purpose for preventing wear. The total indicator run-out reading on the periphery of the chuck, face of body, and face of jaw steps shall not be more than 0.003 inch out of tolerance from a center line of zero. The chuck shall be medium duty and capable of turning the maximum lathe speed without exceeding safety level recommendations. The chuck shall conform to ANSI B5.8.

3.4.10 Chuck, 3-jaw, universal. The chuck body shall be iron alloy (semi-steel) and shall fit the spindle nose without adapters other than as part of the chuck. The chuck shall be universal geared, self-centering, reversible jaw type. The jaws and adjusting screws shall be hardened to a Rockwell scale sufficient for their intended purpose for preventing wear. The total indicator run-out reading on the periphery of the chuck, face of body, and face of jaw steps shall not be more than 0.003 inch out of tolerance from a center line of zero. The chuck shall be medium duty and capable of turning the maximum lathe speed without exceeding safety level recommendations. The chuck shall conform to ANSI B5.8.

3.4.11 Spindle nose collet chuck. The collet chuck shall amount directly on the lathe's spindle nose. It shall be an impact tightening chuck operated by a handwheel and impact mechanism. All parts of the chuck, except the handwheel, shall be of hardened and precision ground alloy steel. The chuck shall accept collets in the increments specified in paragraph 3.4.11.1.

3.4.11.1 Collets. A collet set shall be provided to fit the collet chuck specified in paragraph 3.4.11. The collets shall contain multiple steel jaws permanently bonded in a resilient, non-metallic material which is impervious to high temperature, oil and cutting coolant. The steel jaws in the collets shall be equally spaced to fully grip any diameter work part within a capacity of one-eighth inch for each collet. Collets shall be a complete set to cover the range from 1/16 inch through 1-3/8 inch round stock and shall be boxed in a suitable container.

3.4.12 Stead rest. The steady rest shall be fitted to the ways of the lathe. The steady rest shall be of the hinged type with three adjustable jaws. Adjusting and locking screws for each of the three jaws shall be furnished. Means shall be provided for clamping the steady rest to the bed ways.

3.4.13 Follow rest. One follow rest arranged for quick attachment and removal shall be provided. It shall be equipped with two jaws, each with locking and adjusting screws.

3.4.14 Work light. The work light shall be for mounting on the lathe, have an adjustable or flexible arm, and take a lamp of at least 75 watts.

3.4.15 Cross feed stop. The cross feed stop shall be a single or multiple stop fitted to the cross feed and permit quick retracting of the tool from the cut when threading and return to the previous setting for feeding into the next depth of cut. The cross feed stop shall be designed to permit normal use of the cross feed screw when the stop is not in use.

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3.4.16 Drill chuck. The drill chuck shall be the key type and have a 3/4 inch capacity. Chuck shall be for mounting in the tailstock spindle of the lathe.

3.4.17 Center, live. The live center shall be the ball bearing type for mounting in the tailstock spindle of the lathe.

3.4.18 Dog, set, lathe. The set shall consist of seven lathe dogs of one each in the following size capacity: 1/2 inch, 3/4 inch, 1 inch, 1-1/4 inch, 1-1/2 inch, 1-3/4 inch and 2 inch. The 1-3/4 and 2 inch capacity lathe dogs shall be the clamp type and the 1/2, 3/4, 1, 1-1/4 and 1-1/2 inch capacity lathe dogs shall be the bent tail type for driving work in the lathe.

3.4.19 Holder set, lathe tool, knurling. Knurling tool shall be the self-centering head type with a diamond pattern in fine, medium and coarse knurls. Holder shank size shall be the same as the tool holder size of the tool post furnished with the lathe.

3.4.20 Holder set, lathe tool, cutting-off. The cutting-off tool shall be a set of three tools consisting of one each: straight holder, right-hand offset holder and a left-hand offset holder. Each cutting-off tool shall be complete with a cut-off blade. Holder shank size shall be the same as the tool holder size of the tool post furnished with the lathe.

3.4.21 Holder set, lathe tool, turning. The turning tool shall be a set of three tools consisting of one each; straight holder, right-hand offset holder and a left-hand offset holder. Each turning tool shall be complete with a high speed turning tool bit. Holder shank size shall be the same as the tool holder size of the tool post furnished with the lathe.

3.4.22 Holder lathe tool, boring bar. The boring bar tool shall be a three bar set consisting of one each of the following bar sizes: 1/2 inch diameter by 8 inch length, 3/4 inch diameter by 11 inch length, 1-1/8 inch diameter by 16 inch length. Bars shall be the end cap pattern type complete with 90 degree, 45 degree and 30 degree end caps with tool bits. Holder shall be furnished fitted to the lathe.

3.4.23 Holder, lathe tool, threading. The threading tool shall be a 60 degree sharp V-form cutter. Holder shank size shall be the same as the tool holder size of the tool post furnished with the lathe.

3.5 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat appearance, having the stability and efficient operating characteristics as found in commercial equipment of the type specified herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When specified (see 3.1 and 6.2) first article shall be applied to the preproduction model or initial production item and to the first item produced subsequent to a significant design change in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and the tests in 4.4 and 4.5. Failure of the first article to pass examination inspection or the tests shall be cause for rejection of the unit.

4.2.2 Quality conformance inspection. Unless otherwise specified, each lathe shall be subjected to quality conformance inspection (see 6.2). Quality conformance inspection shall be applied to the production items being offered for acceptance under the contract. Quality conformance inspection shall consist of a. through d. as follows. Failure of an item to pass an examination or test shall be cause for rejection of the unit.

a. Examination (see 4.3).
b. Operational test (see 4.4.1).
c. Alignment accuracy test (see 4.4.2).
d. Packaging inspection (see 4.5).

4.2.2.1 Sampling. When required (see 6.2), sampling for quality conformance inspection shall be in accordance with MIL-STD-105 at the inspection levels and AQL's specified.

4.3 Examination. Lathes shall be examined to determine compliance with section 3 herein and the contract using the following checklist as a guide.

EXAMINATION CHECKLIST

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Safety and health requirements not as specified	3.1.1
Construction not as specified	3.1.3
Cabinet base not as specified	3.1.3.1
Threads not as specified	3.1.8
Lubrication not as specified	3.1.10
Painting not as specified	3.1.12
Dimensions and characteristics not as specified	3.2
Bed and ways not as specified	3.3.1
Headstock not as specified	3.3.2
Headstock spindle not as specified	3.3.3
Carriage not as specified	3.3.4
Cross slide and compound rest not as specified	3.3.5
Cross feed screws not as specified	3.3.6
Apron not as specified	3.3.7
Power feed not as specified	3.3.8
Tailstock not as specified	3.3.9

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EXAMINATION CHECKLIST (continued)

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Data plates not as specified	3.3.10
Electrical system not as specified	3.3.11
Motor not as specified	3.3.12
Lubrication chart or plate not as specified	3.3.13
Nameplate not as specified	3.3.14
Equipment not as specified	3.4 thru 3.4.23
Workmanship not as specified	3.5

4.4 Tests.

4.4.1 Operational test. The lathe shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and other accessories shall be verified during the test period.

4.4.2 Alignment accuracy test. The lathe shall be tested in accordance with the National Machine Tool Builders' Association standard test procedure and the illustrations in ANSI B5.16 to determine compliance with the accuracy requirements of 3.2.1. Failure to meet any of the recommended standards addressed in ANSI B5.16 will be cause for rejection of the lathe.

4.4.3 Performance tests. The lathe shall be subjected to the following turning tests after the machine has reached operating temperature.

4.4.3.1 Round turning test. A bar of low carbon (1020) steel not less than 2 inches in diameter and 4 inches long shall be held in the chuck furnished with the lathe. A rough and finish cut shall be made at least 2 inches long using a cutting speed of not less than 80 feet per minute. The turned diameter shall conform to the accuracy of Test 21, ANSI B5.16.

4.4.3.2 Metal removal turning test. A bar of low carbon (1020) steel approximately 5 inches in diameter and 15 inches long shall be held in the chuck and supported by a tailstock center. A cut shall be made for a length of not less than 12 inches using a single point carbide tip turning tool and a cutting speed of not less than 200 feet per minute. The depth of cut and rate of feed shall be adjusted to produce full load input to the motor. The metal removal rate shall be at least 1 cubic inch per minute per horsepower rating of the motor.

4.4.3.3 Cylindrical turning test. Using the same test bar as for the metal removal test, the specimen shall be semi-finished to the configuration shown in Test 22, ANSI B5.16. A finish cut shall be taken over the two collars in one pass using a single point carbide tip turning tool and a cutting speed not less than 350 feed per minute. The turned diameters shall conform to the accuracy of Test 22, ANSI B5.16. Upon completion of Test 22, the test specimen shall be prepared for turning between centers. A single cut shall be taken over the two collars, and the turned diameters shall conform to the accuracy of Test 23, ANSI B5.16.

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4.5 Packaging inspection. Inspection to determine compliance with cleaning, preservation, packaging, packing, and marking requirements of the applicable packaging documentation, for the level or degree of protection designated in the contract, shall be as specified in section 5.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or Industrial, as specified (see 6.2).

5.1.1 Level A preservation. Level A preservation shall be in accordance with MIL-M-18058.

5.1.1.1 Technical data. Publications issued with the lathe shall be packaged in accordance with method IC-1 of MIL-P-116.

5.1.2 Industrial preservation. The lathe shall be cleaned, preserved and packaged in accordance with the criteria and guidelines of ASTM-D3951-82.

5.2 Packing. Packing shall be level A or B, or Industrial, as specified (see 6.2).

5.2.1 Level A packing. Level A packing shall be in accordance with the requirements of MIL-M-18058.

5.2.2 Level B packing. Level B packing shall be in accordance with the requirements of MIL-M-18058.

5.2.3 Industrial packing. Industrial packing of the lathe shall conform to ASTM-D3951-82.

5.3 Marking.

5.3.1 Military marking. Lathes packaged and packed in accordance with Level A or Level B requirements (see 5.1.1, 5.2.1, and 5.2.2) shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial marking. Lathes packaged and packed in accordance with commercial requirements (see 5.1.2 and 5.2.3) shall be marked in accordance with ASTM-D3951-82.

6 NOTES

6.1 Intended use. This lathe is intended for usage in truck-mounted field repair shops.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. First article, when required (see 3.1).
c. First article inspection, if required (see 4.2.1).
d. Sampling inspection, when required (see 4.2.2.1).
e. Degree of preservation and packaging, and degree of packing required (see 5.1 and 5.2).

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Custodian:
Army - AL

Review interest:
Army - none

User interest:
Army - none

Preparing Activity:
Army - AL

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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